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Multi-Model Gait Fatigue Detection and Feature Sensitivity Analysis Based on the DUO-GAIT Dataset

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***Abstract*—This study focuses on gait-based fatigue detection using the DUO-GAIT dataset. Through a rigorously designed experimental protocol, we conducted a comprehensive performance comparison of five computational models: MLP, BiLSTM, XGBoost, SVM, and Random Forest, complemented by an in-depth sensitivity analysis of feature discriminative power for fatigue classification. The experimental results demonstrate that the BiLSTM model achieves superior performance in fatigue state recognition, with stride times and stride lengths emerging as the most discriminative features for fatigue detection. In addition, ensemble learning models ,such as XGBoost exhibit significant advantages in inference efficiency and model size, making them well-suited for practical deployment. SVM offers a balance between inference speed and model compactness, also showing potential for real-world applications. While deep learning models—especially BiLSTM—excel in recognition accuracy, they are less competitive in terms of inference speed and model size. These findings provide a critical theoretical foundation and practical guidelines for developing wearable device-based gait monitoring systems capable of continuous fatigue assessment in free-living environments. However, this study does not provide an in-depth quantitative analysis of other practical constraints for real-world wearable deployment, such as energy consumption, battery life, device computational resources, or whether the inference latency meets real-time application requirements. Future work will address these aspects to further facilitate efficient deployment of the proposed models on wearable devices.** *(Abstract)*

Keywords-Gait Analysis;Fatigue Detection;Machine Learning; Feature Importance;Deep Learning(key words)

I. Introduction

Gait analysis plays a vital role in sports science, clinical rehabilitation, and movement disorder assessment. Dual-task and fatigue conditions significantly affect gait patterns, and studying these changes helps elucidate neuromotor control. The widespread use of wearable IMUs enables precise gait data collection. The newly released DUO-GAIT dataset [1] offers synchronized multi-modal sensor data under four key conditions: single-task, dual-task, fatigued, and non-fatigued. However, systematic comparisons of machine learning methods and feature importance analyses using this dataset are still lacking. This study conducts comprehensive experiments to support the development of wearable-based gait monitoring systems.

II. Related work

A. DUO-GAIT Dataset Characterization

The DUO-GAIT dataset [1] comprises inertial measurement unit (IMU) recordings from 16 healthy adults (8 male, 8 female) aged 21-35 years. Each participant completed four 6-minute walking trials under controlled experimental conditions: Single-Task Control (ST-Control), Single-Task Fatigue (ST-Fatigue), Dual-Task Control (DT-Control), and Dual-Task Fatigue (DT-Fatigue). A distributed network of nine IMUs sampled triaxial acceleration (±16g) and angular velocity (±1000°/s) at 128Hz from strategic anatomical locations including head, torso, lumbar region, wrists, legs, and feet. The dataset provides multimodal synchronization of: 1) raw kinematic signals, 2) computed spatiotemporal gait parameters, 3) demographic profiles, 4) physiological biomarkers, and 5) cognitive task performance metrics.

B. Current Research Landscape

While IMU-based gait analysis has achieved significant progress[1], [3], three critical limitations persist in fatigue detection research: First, scarcity of publicly available datasets, capturing combined cognitive-motor stressors. Second, insufficient systematic comparison of machine learning architectures for multimodal gait pattern recognition. Third, lack of interpretable feature importance analysis across different fatigue states. Existing approaches often employ suboptimal feature selection strategies and conventional classifiers that fail to fully exploit the temporal dependencies and nonlinear interactions inherent in multimodal gait data.

III. Methodology

A. Data Preprocessing Pipeline

- Data Merging and Label Conversion: Data files from

- both the left and right foot were read separately. For each dataset, a folder label was added to indicate the task type (e.g., single-task, dual-task, fatigue, or non-fatigue condition).
- Feature Extraction and Processing: Relevant features were extracted for each gait cycle.
- Data Type Conversion and Missing Value Imputation: All numerical features were converted to float type, and non-numeric data were coerced accordingly. Missing values were imputed using the mean value of each respective feature to ensure data integrity.
- Normalization:All numerical features were normalized to the [0, 1] range using MinMaxScaler, ensuring comparability across features and preventing differences in scale from affecting model training.
- Label Encoding: The target label was converted to a binary format (0/1) to facilitate subsequent binary classification model training.
- Sliding Window Sequencing: For each subject, data were sorted by stride index. A sliding window [5] of length 10 (window_size=10, step_size=1) was used to generate input sequences, with the label for each sequence assigned as the label of the last stride within the window.
- Dataset Partitioning: To ensure subject independence between the training and test sets, the last three subjects were selected as the test set, while the remaining subjects were used for training and cross-validation.

B. Data Processing Methodology

1) Mean imputation method: A simple and efficient missing value imputation technique. For numerical features (e.g., step length, stride time), the arithmetic mean of the non-missing values is calculated and used to fill in the missing entries.

2) 4-Fold Cross-Validation: The training data is randomly divided into 4 equal subsets. In each iteration, 3 subsets are used as the training set, and the remaining 1 is used as the validation set. This process is repeated 4 times, ensuring that each subset serves as the validation set exactly once. The final validation metric is obtained by averaging the results from the 4 iterations, which helps reduce the variance caused by random partitioning and improves the reliability of model generalization assessment.

C. Model Architecture and Training

Selecting MLP [4], BiLSTM [2], [3], XGBoost, Random Forest [4], and SVM [4], for comprehensive comparison based on their respective strengths in time-series modeling, feature importance analysis, and deployment efficiency. BiLSTM [2], [3] was chosen for its superior ability to capture temporal dependencies in gait data, making it ideal for fatigue recognition tasks. XGBoost and Random Forest were included due to their robustness on tabular data and suitability for resource-constrained environments. SVM [4] was selected for its high reliability and low false positive rates, which are critical in real-world applications. MLP serves as a baseline deep learning model for comparison.

1) MLP: A feedforward neural network with an 11-dimensional input layer corresponding to the raw feature vector. The architecture includes two hidden layers, each with 64 neurons and ReLU activation functions, followed by a 40% dropout layer to mitigate overfitting. The output layer uses a single sigmoid neuron for binary classification. The model is trained using the Adam optimizer with binary cross-entropy loss, 20 epochs, and a batch size of 32.

2) BiLSTM: The architecture consists of two stacked bidirectional LSTM [2], [3] layers: the first with 128 units returning sequences, and the second with 64 units capturing temporal dependencies. Two 40% dropout layers are applied after each LSTM layer to prevent overfitting. The network flattens the output into a 32-neuron fully connected layer with ReLU activation, followed by a sigmoid output layer for binary classification. Inputs are reshaped to (1, 11) to simulate a single time step. Training uses the Adam optimizer (learning rate=0.001), binary cross-entropy loss, 20 epochs, and class weighting.

3) XGBoost : Ensemble decision tree algorithms are used to process raw features, optimizing for log-loss. Models are trained with default hyperparameters, utilizing decision trees' ability to capture complex nonlinear patterns.

*4) RandomForest(RF):*This method builds multiple decision trees and uses voting for classification. With default hyperparameters, it effectively handles complex feature interactions and nonlinear relationships through ensemble decision-making.

5) SVM: A kernel-based classifier with probability estimation enabled. The model uses an RBF kernel . Training is performed with default hyperparameters, except for enabling probability estimates for AUC calculation. change the default, adjust the template as follows.

D. Evaluation Metrics

Model performance is quantified using:

- Accuracy: Overall classification correctness
- Precision: Positive predictive value Recall: True positive rate
- F1-Score: Harmonic mean of precision/recall

Performance visualization includes:

- Curves with AUC computation [3], [4]
- Confusion Matrices detailing class-specific prediction distributions.

E. Feature Sensitivity Analysis

A systematic ablation study quantifies feature importance through controlled feature elimination:

- Establish baseline performance using the complete BiLSTM model
- Train ablated models using individual features [5]

Features are ranked by importance, with results visualized through bar charts to show each feature's relative contribution to fatigue detection.

IV. Experimental Result

A. Model Performance Evaluation Results

Under both single-task and dual-task conditions, performance metrics of the Multilayer Perceptron (MLP), Bidirectional Long Short-Term Memory (BiLSTM) network, XGBoost, Random Forest and SVM models exhibit considerable variability on different measure.

According to Table I, under both single-task and dual-task conditions, deep learning models (Multilayer Perceptron, MLP, and Bidirectional Long Short-Term Memory, BiLSTM) and traditional machine learning models (XGBoost, Random Forest, and SVM) achieve very similar performance across various metrics. For example, MLP and BiLSTM show comparable accuracy and F1 scores to XGBoost and Random Forest, indicating that both deep learning and machine learning approaches can achieve high recognition performance on this dataset.

It is worth noting that deep learning models possess the ability to automatically learn features from raw data, eliminating the need for manual feature engineering. In particular, the BiLSTM model demonstrates superior performance in modeling temporal dependencies. For instance, as shown in Table 1, BiLSTM achieves the best results in the single-task scenario, highlighting its strength in capturing sequential gait features and making it more suitable for time-series data.

In summary, although deep learning and traditional machine learning models exhibit similar overall performance in this experiment, deep learning models—especially BiLSTM—are expected to show greater advantages in more complex tasks or larger-scale datasets due to their self-learning and temporal modeling capabilities.

TABLE I. PERFORMANCE EVALUATION AND AUC COMPARISON ACROSS DIFFERENT MODELS AND CONDITIONS

	Model	Accuracy	Precision	Recall	F1 Score	AUC
DT	MLP	0.993	0.996	0.920	0.956	0.959
DT	BiLSTM	0.993	1.000	0.920	0.958	0.960
DT	XGBoost	0.992	0.981	0.920	0.949	0.987
DT	RF	0.993	1.000	0.920	0.958	0.987
DT	SVM	0.993	1.000	0.920	0.958	0.960
ST	MLP	0.991	0.996	0.895	0.943	0.947
ST	BiLSTM	0.991	1.000	0.895	0.945	0.947
ST	XGBoost	0.992	0.996	0.908	0.950	0.982
ST	RF	0.991	1.000	0.895	0.945	0.980
ST	SVM	0.991	1.000	0.895	0.945	0.947

B. Confusion Matrix Results

The confusion matrix results of each model are shown in Figures 1-5:

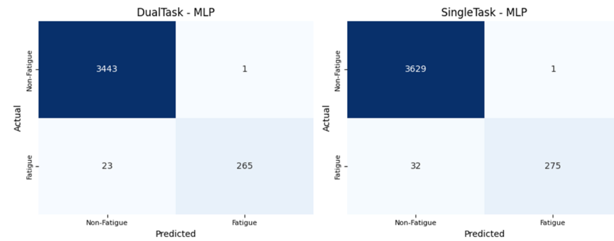

Figure 1. Confusion matrices of the MLP under single-task and dual-task conditio. *(figure caption)*

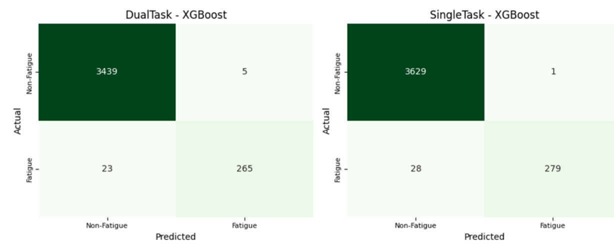

Figure 2. Confusion matrices of the XGBoost under single-task and dual-task condition *(figure caption)*

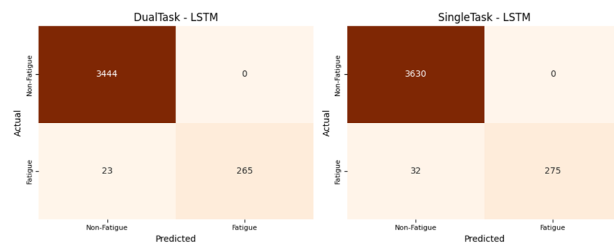

Figure 3. Confusion matrices of the BiLSTM under single-task and dual-task conditio. *(figure caption)*

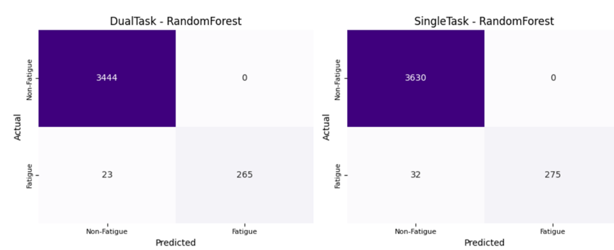

Figure 4. Confusion matrices of the RandomForest under single-task and dual-task conditio. *(figure caption)*

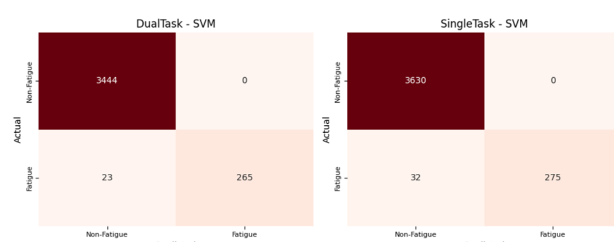

Figure 5. Confusion matrices of the SVM under single-task and dual-task conditio. *(figure caption)*

C. Computational Efficiency And Latency

In terms of inference efficiency and model size, ensemble learning models such as XGBoost and Random Forest demonstrate significant advantages. As shown in Table II, XGBoost achieves inference times of 0.001 ms/sample in both dual-task and single-task scenarios, with model sizes of only 0.166 MB and 0.183 MB, which are substantially smaller than those of deep learning models. Random Forest also offers fast inference times (0.100 ms/sample and 0.001 ms/sample), though its model size is comparatively larger (3.96 MB and 3.82 MB). Deep learning models (MLP and LSTM) are less competitive in both inference time and model size, especially LSTM, which reaches 3.63 MB and inference times around 0.80 ms/sample. SVM falls between ensemble and deep learning models in terms of inference time and model size, with relatively fast inference (about 0.20 ms/sample) and moderate model sizes (2.09 MB and 2.12 MB), making it a viable option for practical deployment.

TABLE II. EVALUATION OF COMPUTATIONAL EFFICIENCY AND LATENCY

	Inference Time (ms/sample)		Model Size (MB)	
	ST	DT	ST	DT
MLP	0.599	0.651	0.314	0.314
BiLSTM	0.802	0.796	3.628	3.628
XGBoost	0.001	0.009	0.183	0.166
SVM	0.200	0.599	2.117	2.092
RF	0.001	0.100	3.824	3.959

D. Feature Sensitivity Analysis Results

Analysis results for Figure 6 are as follows:

1) Spatiotemporal Gait Features Are the Most Sensitive : In the dual-task group, the importance scores for stride_lengths [5] and stride_times are 1.00 and 0.41, respectively, with single-feature model F1 scores of 0.00 and 0.54, both significantly lower than the baseline model (0.91). This indicates that these two features play a critical role in detecting fatigue-related abnormal gait, with stride_time being slightly more sensitive than stride_length. In the single-task group, the importance scores for stride_lengths and stride_times are 0.50 and 0.33, with single-feature model F1 scores of 0.45 and 0.61, showing a similar pattern. This further confirms their strong association with fatigue-induced gait changes, such as reduced stride length and instability.

2) Some Features Are Important but Have Limited Discriminative Power Individually: In the dual-task group, features such as fo_times, ic_times, and clearances_min all have importance scores of 1.00, but their single-feature model F1 scores are 0. This suggests that while these features contribute greatly to the overall model accuracy, they are not effective in identifying fatigue on their own, possibly due to the complex interactions present in fatigue-related gait. Similarly, swing_times has an importance score of 0.67 and a single-feature F1 score of only 0.30, also indicating limited individual discriminative power but substantial impact on the overall model.

3) Limited Contribution of Turning-Related Features: Turning-related features, such as turning_step, have an importance score of -0.0006 in the dual-task group, with a single-feature model F1 score of 0.91, which is close to the baseline, indicating limited discriminative ability for fatigue detection. In the single-task group, the importance score for turning_step is 0, with a single-feature F1 score of 0.91, further demonstrating that turning features contribute little to fatigue detection, which is mainly reflected in changes to straight-walking gait parameters.

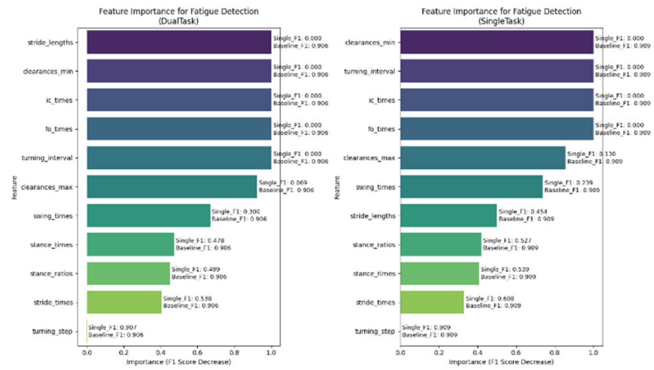

Figure 6. Feature Sensitivity to Fatigue under Single-Task and Dual-Task Conditions Using the BiLSTM Model*(figure caption)*

V. CONCLUSIONS

This study systematically compares the performance of various machine learning and deep learning models for fatigue state recognition. Experimental results show that ensemble XGBoost and Random Forest achieve outstanding AUC, recall, inference speed, and model size, making them highly suitable for deployment on resource-constrained wearable devices. SVM demonstrates balanced performance in terms of precision and model compactness, making it suitable for real-world applications that require high reliability and low false alarm rates. BiLSTM exhibit excellent temporal modeling capabilities and high recognition accuracy in both single-task and multi-task scenarios, making them particularly suitable for system-level applications that demand high accuracy. The MLP struggles to extract multi-level features as task complexity increases, limiting its performance.

In summary, BiLSTM [2], [3] is well-suited for integration into gait monitoring systems to achieve high-accuracy fatigue recognition, while XGBoost is highly suitable for deployment on wearable devices due to its efficient inference and compact model size. Future work will further explore model fusion and feature engineering to improve recognition accuracy and robustness in multi-task scenarios, and will address practical deployment constraints such as energy consumption, device computational resources, and real-time inference requirements through quantitative analysis.

Future work will expand the dataset to include multi-age cohorts (such as adolescents and elderly) and clinical populations. Additionally, we will systematically evaluate and optimize models for deployment on wearable devices, considering constraints like energy consumption, battery life, and computational resources. This includes quantitative analysis of inference latency for real-time monitoring and exploring low-power and hardware adaptation techniques to improve model efficiency and practicality in real-world applications.

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REFERENCES

[1] Zhou, L., Fischer, E., Brahms, C.M., et al. (2023) DUO-GAIT: A gait dataset for walking under dual-task and fatigue conditions with inertial measurement units. Scientific Data, 10: 543.

[2] Hochreiter, S., Schmidhuber, J. (1997) Long Short-Term Memory. Neural Computation, 9(8): 1735–1780.

[3] Aoki, K., Nishikawa, H., Makihara, Y., et al. (2021). Physical Fatigue Detection from Gait Cycles via a Multi - task Recurrent Neural Network IEEE Access, 9: 127565 - 127575.

[4] Kadry, A.M., To Rad, A., Elwan, M.A., et al. (2022). Using Machine Learning to Identify Feelings of Energy and Fatigue in Single - Task Walking Gait: An Exploratory Study. Applied Sciences, 12(6): 3083.

[5] Baghdadi, A., Megahed, F.M., Esfahani, E.T., Cavuoto, L.A. (2018). A machine learning approach to detect changes in gait parameters following a fatiguing occupational task. Ergonomics, 61(8), 1116-1129